A PEDAGOGICAL GUIDE TO EXTENDED PIANO TECHNIQUES

A Monograph
Submitted
to the Temple University Graduate Board

In Partial Fulfillment
Of the Requirements for the Degree of
DOCTOR OF MUSICAL ARTS

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by
Jean-François Proulx
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PREVIEW

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ABSTRACT
A PEDAGOGICAL GUIDE TO EXTENDED PIANO TECHNIQUES
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Extended piano techniques, which mainly involve playing directly on the internal components of the piano, emerged early in the twentieth century, mainly in the United States. Henry Cowell (1897-1965) explored some of these techniques in short piano pieces such as *The Tides of Manaunaun* (1917; clusters) *The Banshee* (1925; glissando, pizzicato) and *Sinister Resonance* (1930; mute, harmonic). Several contemporary composers followed this path. Notably, George Crumb's (b. 1929) mature works combine conventional techniques with an unprecedented variety of extended techniques, including vocal and percussive effects.

Although extended techniques are no longer a novelty, most pianists are still unfamiliar with them. Extended piano works are rarely performed or taught. This situation is regrettable considering the quality of these compositions, and the great potential of extended techniques to expand the piano's coloristic resources. *A Pedagogical Guide to Extended Piano Techniques* is designed to help pianists learn this idiom and achieve fluency. Teachers may also find it useful in planning courses at the undergraduate college level.

Chapters 1 and 2 provide general information such as the development and classification of unconventional techniques, and the construction of the grand piano. Chapters 3, 4 and 5 form the core of the monograph. They contain practical exercises, composed by the author, presented in a progressive order and accompanied by detailed instructions explaining how to execute and practice each technique.

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From a technical point of view, natural string piano techniques—pizzicatos, glissandos, mutes and harmonics—are the basis upon which most other extended techniques are founded. Therefore, this document focuses mainly on mastering these string techniques through the acquisition of two important skills: visual techniques and aural-motor coordination. The following topics are also discussed: use of foreign objects inside the piano, percussive technique, vocal effects, and strategies to integrate various techniques into a unified musical context. Prepared piano techniques are excluded because they do not require the acquisition of performance skills, which form the essence of this monograph.

Chapter 6 provides a suggested course schedule to organize the pianist's study and practice time, and offers practical advice for the preparation of performances involving extended piano techniques. It is hoped that this document will play a positive role in helping pianists gain the knowledge, ability, confidence and enthusiasm to explore and perform extended piano repertoire.

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My parents, godparents, grand-parents, brother and sister, who gave me unconditional love and support since I was born, and without whom I would not be here today.

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To my mother, Nicole, and my father, Carol, with all my love

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CHAPTER 1

INTRODUCTION

Present Situation

In the last 50 years, a good number of composers have integrated extended techniques into their piano writing, at least on a basic level.[1] A few pianists have invested time in mastering and performing this repertoire with great talent. In spite of that, one hears this repertoire only on rare occasions. This is due, in part, to the difficulty of learning extended techniques, and to the difficulty of projecting these delicate sounds effectively to an audience. It is fair to say that extended piano techniques still represent an obscure branch of contemporary music, which is itself obscure and unpopular for a majority of music amateurs. This is regrettable considering the energy and creativity deployed by the outstanding musicians who have opened these new paths.

The marginal status of extended piano techniques is also self-perpetuating: most composers and pianists have not had the chance to hear impressive performances involving these techniques, which leads them to assume that it is not worth the effort to study them seriously. As a result, there is limited information about extended piano techniques available for students, whether in the form of writings or through their teachers. There are three main sources of information for students and pianists who want to learn about this topic: composers who use these techniques—they often write detailed instructions about the execution of special techniques in their scores, and it is sometimes possible to contact them directly if they are living;

[1] Works of some composers are discussed in the appendix section of this document.

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professional pianists who have performed extended repertoire; and a handful of doctoral theses and dissertations—the most complete and useful of which is *Off Key: A Comprehensive Guide to Unconventional Piano Techniques*, by Laurie Marie Hudicek. These sources can be reliable but are sometimes difficult to access, contradictory, and not comprehensive. Pianists still need to discover many things by themselves, often by trial and error.

The *New Instrumentation* Series is a collection of books published by the University of California Press and written by professional instrumentalists of different disciplines—contrabass, violin, guitar, pedal harp, flute, oboe, clarinet, trombone, voice—about the execution of extended techniques. These manuals are addressed both to performers and composers, and explore in depth the extended possibilities of each respective instrument. Unfortunately, the piano is excluded from this series, which is surprising considering the importance of the extended piano in contemporary repertoire.

The acquisition of basic extended technique skills should be part of the musical education of piano students. Among other benefits, it could help them to develop a sensitive ear for tone colors, dynamic levels and balance. These qualities are useful even in traditional piano playing and chamber music. Students who develop a strong affinity for extended techniques could then delve deeper into this subject if they wish.

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Purpose and Content of this Monograph

This document provides pianists with progressive exercises, information and a methodology to learn extended techniques and achieve fluency. These exercises are somewhat similar to the ones found in *The Virtuoso Pianist in 60 exercises* by Charles-Louis Hanon or *51 Exercises* by Johannes Brahms. While some pianists prefer to acquire new techniques simply by

learning the repertoire containing these techniques, preparatory exercises often prove more effective in this regard. It is far easier to overcome a difficulty when it is isolated than when it is included in a complex musical context. Furthermore, technically proficient pianists can make difficult passages seem easy because they have previously practiced and mastered exercises that are even more demanding. This allows them to focus on the musical elements of a piece rather than struggling with its physical execution.

Because extended techniques are still obscure for many, Chapter 2 introduces some basic concepts and information essential to the study of these exercises. In Chapter 3, elementary exercises are accompanied by descriptions and suggestions on how to execute them. The objective is to enable the pianist to play directly on the piano strings and become more familiar with the inside of the piano. As the pianist learns and masters the first exercises, more advanced ones are introduced, building upon the preceding ones. The exercises in Chapter 4 not only increase in difficulty and variety but also require the use of some objects. While it is impractical to examine all the objects a composer could possibly use in any given work, the examples offered in this document can help the pianist to prepare for most eventualities. Chapter 5 contains exercises that combine various extended and traditional techniques. The goal of this chapter is to develop physical coordination as well as the ability to balance different sounds and integrate them into a musical context. In conclusion, Chapter 6 offers practical advice to help one organize practice time, prepare for performances involving extended techniques, and solve potential problems with confidence and authority.

It is important to note that prepared piano techniques are excluded from this document. Preparing the piano is a *passive* musical technique because it is not part of the performance; it

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does not require skills such as coordination, rapidity, or timing, which form the essence of this document. In addition, this topic is vast enough to be the focus of another study.

By definition, *extended* techniques are an extension of regular piano technique. Trying to learn these challenging techniques without a solid musical foundation is not likely to yield satisfying results. For this reason, this document is addressed to pianists who have achieved at least an early advanced level, equivalent to that of an undergraduate Music major.

Pianists who are interested in acquiring these skills might become discouraged in their attempts by lack of a clear starting point, precise goals, and direction. It is hoped that this document will guide the pianist from point zero to a satisfying level of proficiency, while keeping his or her interest throughout. It is also hoped that teachers will find this document useful in planning individual lessons or seminars concerned with these techniques, and possibly composers too, in understanding better the extended possibilities of the piano.

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Pioneers

The explorations that gave birth to extended piano techniques in the early twentieth century represent the culmination of an increasing interest in timbre. Concerns such as how timbres can be combined, modified, and utilized to create musical atmospheres and feelings, while incidental in the Baroque era, grew gradually in importance and have come to the foreground in contemporary music.

At the turn of the twentieth century, composers explored an incredible number of new paths, creating an explosion of musical ideas and styles. According to certain sources,[2] it is around that time, in this feverish artistic climate, that some avant-garde pianists first started

[2] Mainly: Michael Hicks, *Henry Cowell, Bohemian.* (Urbana: University of Illinois Press, 2002).

reaching directly for the strings in their improvisations. Modifying and extending the natural sound of the instrument by interfering with the normal functioning of its playing mechanism was an important step in establishing the primal role of timbre in music, as far as the acoustic piano is concerned. Percy Grainger's *In a Nutshell Suite* (1916) is the first published piano composition to include a passage that must be played directly on the strings with a mallet. However, the real explorer in this field was the American composer Henry Cowell (1897-1965). Within his works, he provided a fascinating catalog of techniques to produce new timbres on the piano: harp-like plucking and strumming, sustained string sonorities, muted tones, harmonics, and clusters are the most important. With the exception of *The Banshee* (1925), the value of his compositions is not universally recognized. However, Cowell's exceptional contribution was to be taken over by other pioneers who pondered the possibilities of extended techniques and developed this idiom into a worthwhile, new musical tradition.

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Special keyboard techniques

Some unconventional piano techniques are not considered to be "extended" because they do not alter the natural timbre of the piano. The *tone cluster* is the most important. It is an amalgam composed of at least three consecutive notes (for example, C-D-Eb) played simultaneously as a chord. The cluster is not used for its harmonic implications, but for its sonority. On the piano, one can produce a white-key cluster, a black-key cluster, or a mixed (chromatic) one. Clusters can be played with the fingers, the fist, the hand, or even the arm, depending on how many notes are to be included. This technique appeared in various compositions early in the twentieth century. Charles Ives' *Scherzo: Over the Pavements* (1907, revised 1913) is a chamber work written for nine performers in which the piano part contains

tone clusters. Humfrey Anger used tone clusters in his piano piece *Tintamarre* (1911) to imitate the sound of bells.[3] In the 1910's, pianist Leo Ornstein often performed his piece *Wild Men's Dance* (1914) which contains an abundance of tone clusters. Some of Henry Cowell's cluster pieces such as *Adventures in Harmony* (1913), *Dynamic Motions* (1916), *The Tides of Manaunaun* (1917) and *Tiger* (1930) gave him the reputation of being the loudest pianist in the world. He was the musician who made the most effort to promote this technique as a legitimate musical device.

Another important special keyboard technique, the *sympathetic string resonance,* is done by depressing a key or group of keys silently and holding them down. The dampers of these silent keys being raised let their related strings vibrate in reaction to the vibrations of other active notes. Surprisingly, the first published example of this technique can be found in Robert Schumann's *Carnaval* Opus 9 (1835). The movement entitled *Paganini* contains an Eb^{7} chord, which must be played by depressing the notes silently. These notes then emerge *ppp* from the sonority of the previous *ff* chords when those are cleared with the pedal.

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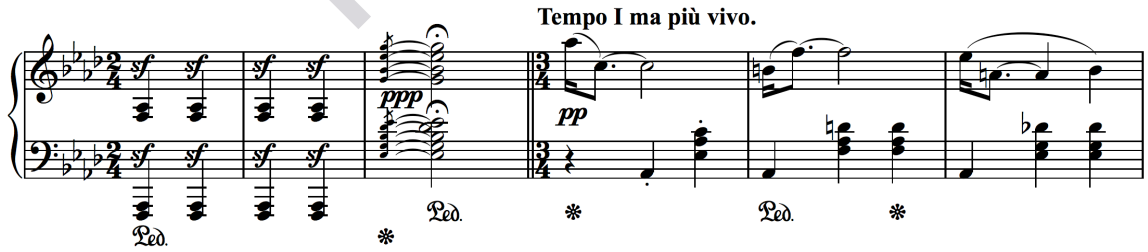
Example 1. Robert Schumann, *Paganini* (from *Carnaval* Opus 9), mm. 35 to 40.



[3] William Lock, *Anger, Humfrey* [on-line] available from http://www.thecanadianencyclopedia.com; Internet, accessed 29 April 2009.